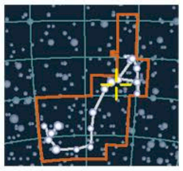
GLOBULAR CLUSTER

M4



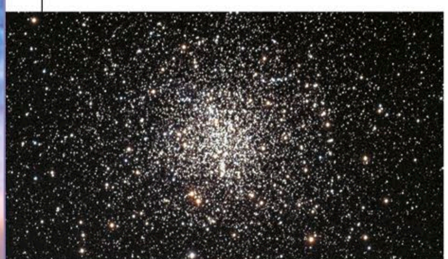
CATALOG NUMBERS M4, NGC 6121

DISTANCE FROM SUN 7,000 light-years

MAGNITUDE 7.1

SCORPIUS

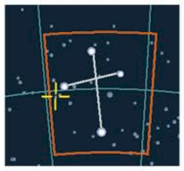
M4 is one of the closest globular clusters to Earth and can be seen with the naked eye on a dark, clear night. The cluster has a diameter of about 70 light-years and contains more than 100,000 stars, but about half the cluster's mass resides within 8 light-years of its center. The Hubble Space Telescope has revealed a planet within M4, with about twice the mass of Jupiter, orbiting a white dwarf star. The planet is estimated to be 13 billion years old.



DENSE CENTER

OPEN CLUSTER

Jewel Box



CATALOG NUMBER NGC 4755

DISTANCE FROM SUN 8,150 light-years

MAGNITUDE 4.2

CRUX

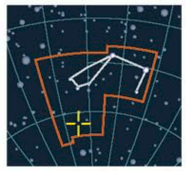
GLITTERING JEWELS

The Jewel Box, also known as the Kappa Crucis Cluster, is an open cluster of about 100 stars and is about 20 light-years across. At less than 10 million years old, it is one of the youngest open clusters known. The three brightest stars are blue giants, while the fourth-brightest star is a red supergiant. The different colors are very apparent in photographs of the cluster, hence its popular name. Lying within the constellation Crux, the Jewel Box is visible only to observers in the Southern Hemisphere.

SOUTHERN SPECTACLE
An optical image (top) captures 47 Tucanae and the Small Magellanic Cloud (see p.311), a satellite galaxy of the Milky Way. A close-up of 47 Tucanae (boxed) reveals one of the most spectacular globular clusters in the sky.

GLOBULAR CLUSTER

47 Tucanae



CATALOG NUMBER NGC 104

DISTANCE FROM SUN 13,400 light-years

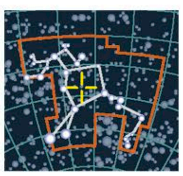
MAGNITUDE 4.9

TUCANA

47 Tucanae is so named because it was originally cataloged as a star—the 47th in order of right ascension in the constellation Tucana. In reality, it is the second-largest and second-brightest globular cluster in the sky, containing several million stars—enough to make a small galaxy. These stars are spread over an area about 120 light-years across, and the cluster's central region is so crowded that there is a high rate of stellar collisions. As a globular cluster ages, the stars within it also age, but 47 Tucanae is home to a number of blue stragglers—stars that are too blue and too massive to still be there if they were original members of the cluster. Astronomers have determined that it is the stellar collisions within the cluster that cause the formation of these blue stragglers.

GLOBULAR CLUSTER

Omega Centauri



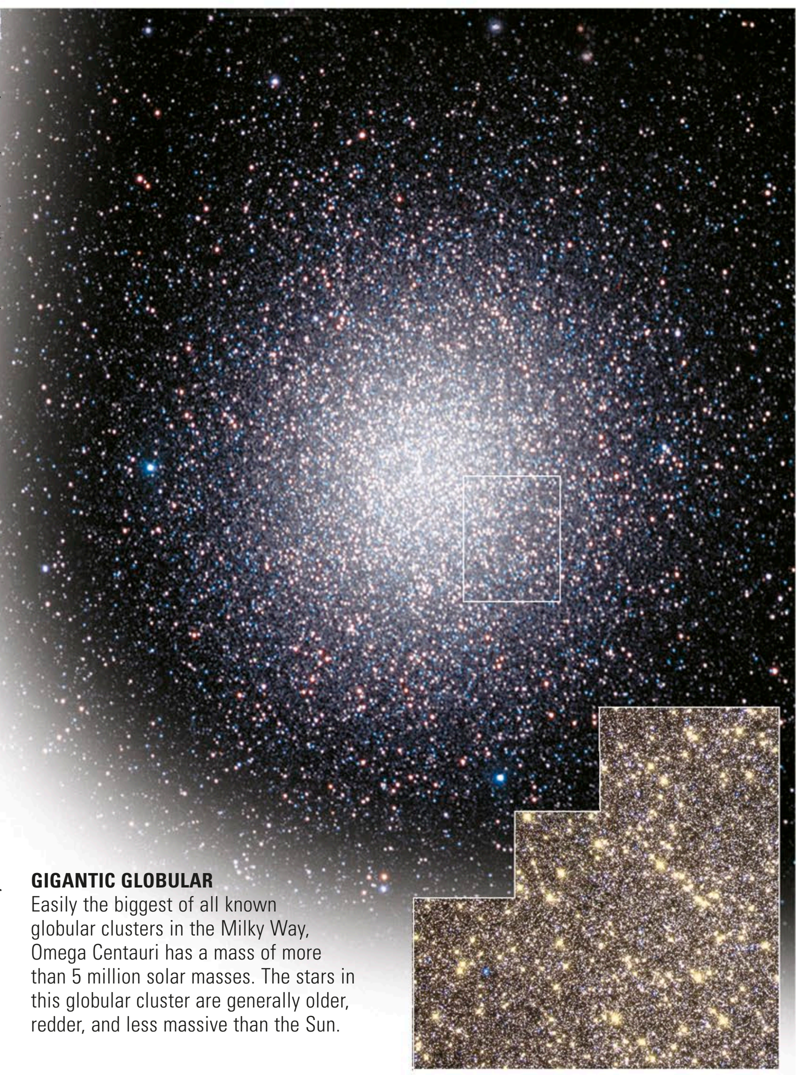
CATALOG NUMBER NGC 5139

DISTANCE FROM SUN 17,000 light-years

MAGNITUDE 5.33

CENTAURUS

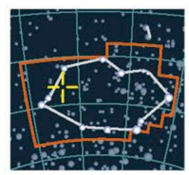
Omega Centauri is the largest globular cluster in the Milky Way—up to 10 times as massive as other globular clusters. Containing more than 10 million stars and having a width of 150 light-years, Omega Centauri is as massive as some small galaxies. To the naked eye, it appears as a fuzzy star, but a small telescope starts to resolve its individual stars. Studies of the cluster's stellar population have revealed that Omega Centauri is one of the oldest objects in the Milky Way—almost as old as the universe itself—and that there have been several episodes of star formation within the cluster. This is unusual for a globular cluster, and one explanation for this is that Omega Centauri may once have been a dwarf galaxy that collided with our own. It would have had about 1,000 times its current mass, but the Milky Way would have ripped it apart, leaving Omega Centauri as the remnant core.



GIGANTIC GLOBULAR
Easily the biggest of all known globular clusters in the Milky Way, Omega Centauri has a mass of more than 5 million solar masses. The stars in this globular cluster are generally older, redder, and less massive than the Sun.

GLOBULAR CLUSTER

NGC 3201



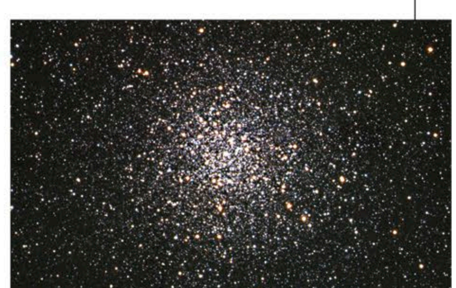
CATALOG NUMBER NGC 3201

DISTANCE FROM SUN 15,000 light-years

MAGNITUDE 8.2

VELA

The globular cluster NGC 3201 contains many bright red giant stars, which give the cluster an overall reddish appearance. The cluster lies close to the galactic plane, so its appearance is further reddened by interstellar absorption. With a visual magnitude of only 8.2, the cluster is too faint to be seen with the naked eye. NGC 3201 is less condensed than most globular clusters, and several observers have suggested that some of the stars appear in short, curved rays, like jets of water from a fountain.



RED-TINGED CLUSTER